

Gabe Flanagin

flanagg@wwu.edu

Research Interests

Partial Differential Equations · Mathematical Optimization · Numerical Analysis · Scientific Computing

Education

Western Washington University

Bellingham, WA

M.S. in Mathematics

2025–Present

B.S. in Mathematics

2021–2025

Research Experience

Graduate-level Research in Industrial Projects for Students (G-RIPS)

Summer 2026

Zuse Institute Berlin (ZIB)

- Fine-tuned task-specific foundation models for cryo-electron tomography.
- Trained models to segment biological structures in 3D tomograms, dealing with issues imaging challenges such as low noise-to-signal ratios and missing wedge.

Terrain-Aware Reaction–Diffusion Models & Optimal Control

2026–Present

Collaborator: Alex Escoto

- Developing reaction–diffusion models for infectious disease spread in wildlife populations with terrain-influenced movement.
- Designing computational methods for domains with complex topography and heterogeneous land cover.
- Investigating optimal control strategies including culling, vaccination, and predator introduction.

Boundary Integral Methods for Interface Problems

2026–Present

Advisor: Dr. Zhen Chao

- Investigating boundary integral formulations for elliptic interface problems.
- Applications to electrostatic models and PDEs on complex geometries.
- Implementing numerical methods, validating formulations, and performing convergence analysis.

Using Max-Plus Algebra to Compute Reachability in Dynamic Systems

2025

Advisor: Dr. Rick Barnard

- Implemented a max-plus algebra algorithm for approximating Hamilton–Jacobi–Bellman equations.
- Proved applicability of the method to a broader class of functions than previously demonstrated.
- Generalized semiconvexity results underlying the approach.

Publications

[1] **Flanagin, G.**, & Berger, R. (2026).

Rationalizing Patterns in the Cation Ordering, Geometric Distortion, and Electronic Properties of a Class of I–V–VI₂ Chalcogenide Semiconductors.

Journal of Physical Chemistry C.

DOI: [10.1021/acs.jpcc.5c05853](https://doi.org/10.1021/acs.jpcc.5c05853)

Presentations & Talks

- *An Introduction to Spatiotemporal Rabies Models.* WWU Mathematics Colloquium Series, April 2026.
- *Using Max-Plus Algebra to Compute Reachability in Dynamic Systems.* WWU Mathematics Scholars Week, May 2025.

Teaching Experience

Western Washington University

Bellingham, WA

Graduate Teaching Assistant

2025–2026

- Teaching Assistant — MATH 115: Precalculus II Spring 2026
- TA Liaison — MATH 114: Precalculus I Winter 2026
- Teaching Assistant — MATH 112: Functions & Algebraic Methods Fall 2025

Programming & Computational Tools

Python (NumPy, SciPy), FEniCS, PyTorch, Linux, CUDA, Slurm, HTCondor, Git, MATLAB